Calibrated Convergence Control for KKR / BdG Green's-Function DFT

An AiiDA-native early-abort framework (`ml_assist`), a live pilot, and a held-out benchmark

Prepared for collaboration hand-off

2026-07-21

Abstract

Self-consistent-field (SCF) convergence is the practical bottleneck of KKR and Bogoliubov–de Gennes (BdG) Green's-function DFT campaigns: many runs stall or diverge, wasting cluster time. We present an **AiiDA-native convergence-control framework**, `ml_assist`, that scores a run's first ten SCF iterations and can abort doomed runs early — an opt-in layer with a zero-risk default. The framework ships a **frozen `numpy` random-forest scorer** (the AiiDA daemon has no scikit-learn) alongside a **simple RMS-at-iteration-10 baseline**, both wrapped in a **per-cohort conformal early-abort policy** that controls the false-abort rate after per-cohort calibration under calibration/test exchangeability, with Mondrian per-stratum thresholds for conditional coverage. A **live Fe-doped $NbSe_2$ pilot validates the abort machinery end-to-end**: a true early abort saved ∼130 SCF iterations against its diverging counterfactual, and a would-be false abort on a slowly-converging run was correctly withheld (N=5, descriptive). On a **336-run, leakage-clean held-out replay with per-run convergence-target (`QBOUND`) labels**, the learned model does *not* beat the RMS@10 baseline on the primary $NbSe_2$ normal-state cohort (held-out AUC 0.83 vs. 0.90), *does not transfer across materials* (other-material AUC ≈ chance), and a $NbSe_2$-seeded conformal threshold does not transport (realized false-abort 0.47) — though **per-cohort recalibration restores** realized ≈ target α. We conclude that *reliable deployment requires per-cohort calibration and strong simple baselines*; the honest, provenance-tracked framework — not model superiority — is the contribution. Separately, an engineering effort repaired four cell-inappropriate BdG-workchain defaults, enabling the first end-to-end multi-atom BdG run and its memory-scaling number. Everything ships as an installable `aiida-kkr-mlassist` extension with unit tests and pre-registered success gates.

Contents

1 Goals and scope 1

2 Background and tooling (what we used and why) 2

3 Data and the provenance-mining audit 2

4 The convergence classifier 3

5 The held-out benchmark: does the model beat a simple RMS rule? 4

6 The conformal early-abort policy 5

7 Deployment evaluation (per-cohort, calibrated) 7

8 Live pilot: the abort machinery works end-to-end 8

9 The BdG Δ-channel 9

10 Workflow hardening: four BdG footguns and the memory number 9

11 Software package 10

12 Pre-registration and success gates (superseded) 11

13 Side studies and honest negatives 11

14 Reproducibility and governance 11

15 PotentialBank v1: L3 warm-start realized 12

16 Status and roadmap 13

A Result tables 13

B Model provenance 13

1 Goals and scope

The parent project is a multi-year KKR / KKR-BdG superconductivity effort on 2H-$NbSe_2$ with transition-metal dopants (V/Mn/Fe/Co), plus a Bi/NbBi subset. SCF convergence is the practical bottleneck: many runs stall, diverge, or converge slowly, wasting cluster time. `ml_assist` targets three *acceleration levers*:

L1 — early abort stop runs that are predicted not to converge, recovering their wasted tail.

L2 — probe-and-commit mixing race a few mixing settings briefly, commit the best.

L3 — warm start reuse a converged potential to halve iterations.

This report covers the **v1 deliverable (L1)**, fully built and validated, plus the infrastructure (feature schema, frozen models, conformal policy, package, workchain hardening) that L2/L3 reuse. **L3 (warm start) is now realized as PotentialBank v1** (Sec. 15): a certified catalog of single-structure converged potentials with measured same-structure warm-start savings.

Design principles. (1) *Opt-in, zero-risk default*: absent an `ml_assist` input the workchain behaves identically to stock. (2) *Advisory by default*; abort only when explicitly enabled and the statistical guarantee is certifiable. (3) *Per-system deployment*: we do not claim cross-material transfer. (4) *Honesty first*: negative results are reported as-is and are publishable.

2 Background and tooling (what we used and why)

JuKKR (KKRhost, voronoi, KKRimp). The Green's-function DFT engine. SCF proceeds by iterating the potential; each iteration reports an RMS error and charge-neutrality. *Why it* *matters for ML:* these per-iteration quantities are the raw signal from which convergence fate is predicted.

BdG extension. Superconducting pairing is treated by doubling the Hamiltonian into Nambu (particle–hole) space; KKRhost-BdG additionally prints a pairing amplitude Δ per iteration. *Why:* the pairing channel is a BdG-specific failure mode invisible to normal-state features.

AiiDA 2.7. Workflow engine and provenance store. All calculations, inputs, and outputs are nodes in a directed provenance graph. *Why:* it gave us a mineable corpus of historical runs *and* is the natural place to deploy an opt-in wrapper workchain with full audit trail.

aiida-kkr. The AiiDA plugin providing `kkr_scf_wc` (normal SCF) and `kkr_bdg_wc` (BdG chain). *Why:* `ml_assist` subclasses `kkr_scf_wc` to add the prediction hook.

scikit-learn (training only). Random forests for the classifier. *Why RF:* in distribution it beats logistic regression and the physical extrapolation heuristic and needs no feature scaling (Table 3). *But out-of-sample* the strong baseline to beat is a plain RMS-at-iteration-10 rule, which the model does *not* beat on the primary cohort (Sec. 5). *Constraint:* scikit-learn is *not* installed in the AiiDA daemon environment.

numpy (inference). We export the trained forest to a flattened-array `.npz` and traverse it in pure `numpy`, reproducing scikit-learn's `predict_proba` exactly. *Why:* the daemon must score runs without scikit-learn; a numpy-only path removes that dependency (Sec. 4).

Conformal prediction. A distribution-free wrapper that turns a score into a decision with a controlled error rate. *Why:* it lets us *control the false-abort rate* $\leq \alpha$ after per-cohort calibration under calibration/test exchangeability, regardless of the model's calibration — the safety property an abort policy needs (Sec. 6).

3 Data and the provenance-mining audit

The AiiDA database holds ∼7,700 calculation nodes and ∼6,400 workchains (2021–2026). We mined per-iteration RMS/charge-neutrality trajectories and outcomes. Three *leakage-class* corrections were essential and define the trustworthiness of every downstream number:

1. **Segment de-inflation.** Training rows were KKR *segments* (up to 5 per workchain), not runs. Deduplicating to one cold-start row per run reduced 3,405 segments to **2,710 runs** (1,851 normal, 90% converged).[1]
2. **Censoring scope.** 389 excepted/killed calculations are 100% censored; the dataset is *completed runs only.* Crashes/NaN-kills are out of scope and handled by a watchdog, not the model.
3. **The Replay-OOF Invariant.** Any replay, preview, or "deployment simulation" number is out-of-sample or prospective *by construction*, never in-sample. This was adopted after an in-sample shadow replay produced a spurious recall of 1.000; the honest out-of-sample number is 0.70 (Sec. 7). The model that scores a run never trained on it (verified by node id).

These corrections deflated some earlier claims (e.g. a cross-material transfer number collapsed from 0.94 to chance) — an outcome we treat as a contribution, not an embarrassment.

[1] Modeling uses the 1,830 normal runs with at least $T_{\text{obs}}+2$ iterations (21 too-short runs excluded, needed to form the observation window); the BdG Δ study uses the 641 of 846 BdG runs whose `out_kkr` was retrievable and parseable (Sec. 9).

4 The convergence classifier

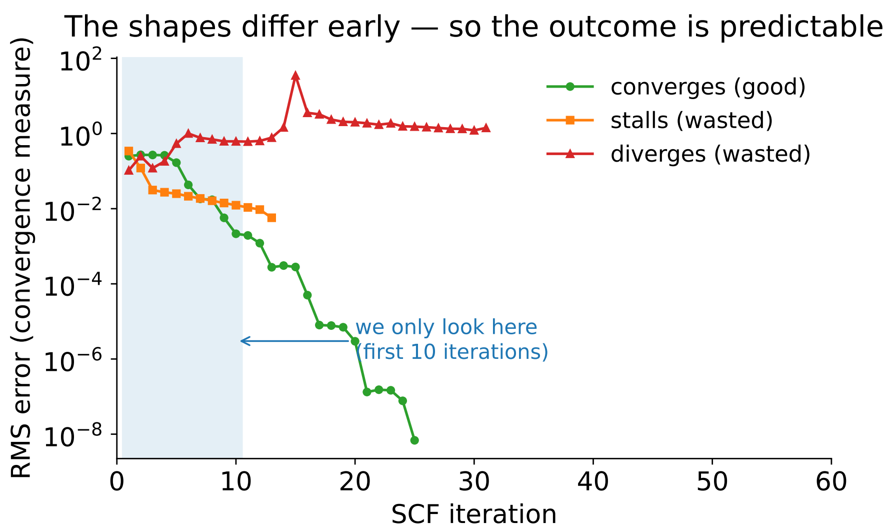

Figure 1: Why early prediction is possible. Converging, stalling, and diverging runs already trace distinct shapes within the first ten iterations (shaded) — the only data the model sees.

Features. A 15-dimensional vector computed from the first $T_{\text{obs}} = 10$ iterations: log-RMS window statistics (first, last, mean, std, min, drop, slope), two charge-neutrality statistics, and three mixing parameters each with a missing-flag. A single canonical implementation is shared by training and inference so they cannot drift (verified bit-identical).

Model and in-distribution accuracy. A 400-tree balanced random forest. Its *in-distribution* 5-fold cross-validation AUC is strong where data is plentiful and honestly weaker where it is thin (Table 2); the pooled normal-state 5-fold AUC is 0.944. In distribution it beats logistic regression and a physics extrapolation heuristic (Fig. 2, Table 3). **We deliberately do *not* treat 0.944 as the deployment headline**: it is a same-distribution cross-validation number. The deployment-relevant question — how the frozen model scores runs it has never seen, against a simple RMS baseline — is answered by the leakage-clean held-out benchmark of Sec. 5, and there the model does *not* beat RMS@10 and does *not* transfer across materials. The model is therefore deployed per-system with per-cohort calibration.

A dropped complication (reported for the record). A schedule-aligned observation window was a pre-registered design lock intended to protect "flat-start" runs (flat, high-RMS early, then converging). On the deduplicated data it gave *no* measurable benefit: the fixed window already separates flat-starters at AUC ∼0.90 (the discriminative signal is in charge-neutrality and mixing, not the flat RMS slope). It was dropped for v1; the frozen model uses the plain fixed window.

Pure-`numpy` export and the float32 gotcha. The forest is flattened to node arrays (children, feature, threshold, per-leaf class probabilities) and traversed in `numpy`. Golden-parity against scikit-learn is **exactly** 0.0, verified running under the daemon interpreter (numpy 1.26, no scikit-learn). One subtlety was decisive: scikit-learn casts inputs to `float32` for tree traversal, so a `float64`

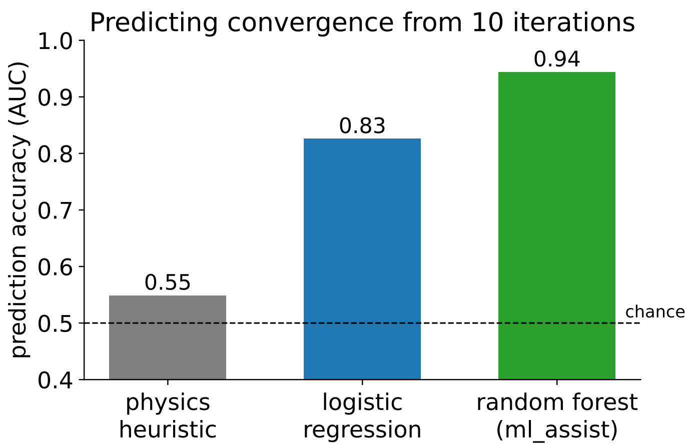

Figure 2: *In-distribution only.* From ten iterations, the random forest cross-validates better than a physics extrapolation heuristic or logistic regression (AUC; 0.5 is a coin flip). This is *not* the deployment metric — see the held-out benchmark (Fig. 3), where a simple RMS@10 rule matches or beats the model out-of-sample.

comparison mis-routes samples sitting on a threshold tie (an initial 10^{-2} parity failure). The numpy path compares in `float32`; embedded golden vectors (including the exact edge sample that exposed the bug) guard against regression.

5 The held-out benchmark: does the model beat a simple RMS rule?

The honest test of deployment value is a *leakage-clean, out-of-sample* evaluation with correct labels. We assembled **336 completed iffslurm runs verified absent from training** (by node id) and scored each run's frozen first-ten-iteration features. Three corrections over an earlier pooled analysis were decisive (they revise our own earlier, over-optimistic numbers):

1. **Per-run `QBOUND` labels.** Convergence is judged against each run's own convergence target, not a fixed 10^{-3}; a fixed cut mislabelled 57 runs (many $NbSe_2$ runs use `QBOUND`$= 0.008$).
2. **Censoring.** 15 excepted/killed runs (exit $\neq 0$) are dropped from all denominators.
3. **No pooled headline.** Numbers are reported *per cohort*; a single pooled AUC conflates a warm/restart population (which mostly converges) with cold starts (which mostly fail) and is not a deployment metric.

The result (Table 1, Fig. 3). On the primary target — $NbSe_2$-family normal-state — the frozen model reaches held-out AUC **0.83** [0.77, 0.89] (cold starts 0.79), while a **conformally-calibrated RMS@10 rule reaches** 0.90 [0.85, 0.94] and gives *higher* failed-run recall and compute savings at matched false-abort at every α (Sec. 7). **The learned model does not beat the simple RMS baseline on the primary cohort.** It is anti-predictive across materials (other-KKR AUC 0.48; a warm/restart cohort 0.35) — a genuine transfer failure, since RMS@10 remains predictive on the same runs (0.90, 0.61). The one cohort where the model edges RMS@10 ("unknown", 0.80 vs. 0.78) is small (n=42). Excluding the 56 borderline runs (final RMS within a factor 3 of `QBOUND`) leaves the ranking unchanged.

Table 1: Held-out benchmark, per cohort (leakage-clean, per-run `QBOUND` labels, censored dropped). AUC [95% bootstrap CI]. *No pooled headline number is reported.* A conformally-calibrated RMS@10 rule matches or beats the frozen model everywhere except the small "unknown" cohort; the model is anti-predictive across materials (other-KKR, warm/restart).

cohort	n	conv.	frozen ML	RMS@10	RMS-slope
$NbSe_2$-normal (primary, all)	167	38	0.831 [.77,.89]	**0.897 [.85,.94]**	0.740
— cold starts	128	12	0.785 [.68,.88]	**0.901 [.80,.97]**	0.568
— warm/restart	39	26	0.430	0.414	0.467
Bi/NbBi	28	21	0.898	**0.939**	0.816
other-KKR (cross-material)	45	34	**0.476**	0.904	0.583
BdG-misfit (normal model on BdG)	78	41	0.581	0.813	0.631
unknown	42	10	**0.800**	0.778	0.878
warm/restart-like (all families)	93	67	**0.350**	0.606	0.738

Scope consequences. (i) BdG runs must be scored with the BdG model, not the normal one (normal-model-on-BdG AUC 0.58). (ii) Warm/restart runs are *outside the current abort policy*: their failures are rare and not separable from the early window (both methods ≈ chance), so they are excluded from abort and left to run. (iii) No cross-material transfer is claimed for the learned model.

6 The conformal early-abort policy

Guarantee. Given a target α, we abort a run iff its predicted convergence probability is below a threshold calibrated on the campaign's converged runs. One-sided split conformal makes

$$\Pr(\text{abort} \mid \text{the run would actually converge}) \leq \alpha.$$

Certifiability needs ∼19 calibration points at $\alpha = 0.05$ (∼9 at 0.10); a campaign seeds these from material-matched history and refreshes online. The guarantee holds under *exchangeability* of calibration and test runs; material-matched seeding plus online refresh is how we keep that assumption approximately valid as a campaign drifts; a pre-registered violation trip-wire reverts to advisory mode if the realized false-abort rate ever exceeds the budget.

The marginal-vs-conditional gap (found and fixed). The split-conformal guarantee is *marginal*. Verified in out-of-sample replay: a single (marginal) threshold gives overall false-abort 0.049 at $\alpha = 0.05$ but **0.086 conditional on the flat-starter stratum** — the very runs the dropped aligned window was meant to protect. The fix is **Mondrian (group-conditional) thresholds**: one threshold per stratum (flat-starter vs. other, derived directly from the feature vector, agreeing 100% with the trajectory definition). This restores conditional control to **0.047** (Fig. 4, Table 4). Both vehicles deploy Mondrian.

Seeded thresholds do not transport — recalibrate per cohort. The guarantee holds only under *exchangeability* of calibration and test runs, and the held-out data show this is not automatic. A threshold seeded on historical $NbSe_2$ converged scores, applied to the broader held-out cohort, gave a realized false-abort rate of **0.47** at $\alpha = 0.10$ — far above target: the seed does *not* transport across cohorts. The fix is **per-cohort recalibration**: re-deriving the threshold on the deployment

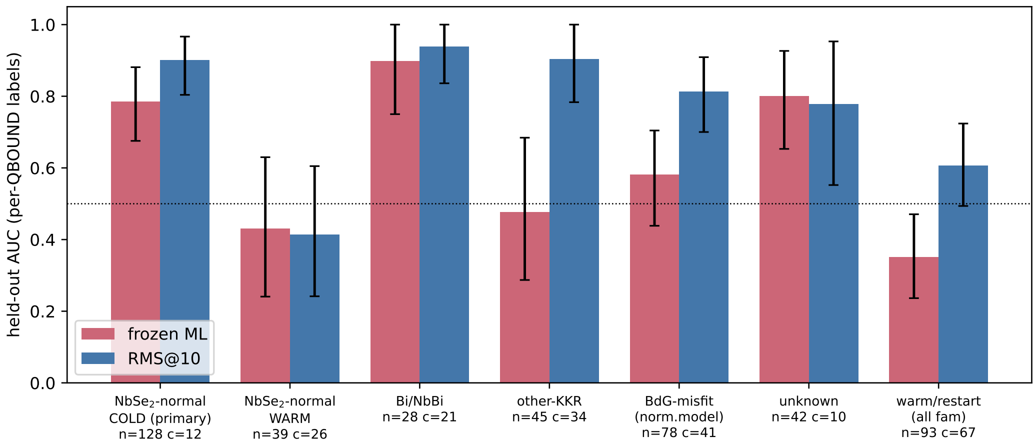

Figure 3: Held-out per-cohort AUC (per-run `QBOUND` labels), frozen ML vs. RMS@10, with 95% bootstrap CIs. On the deployment target ($NbSe_2$-normal cold, leftmost) RMS@10 leads; the model is at or below chance on other-material and warm/restart cohorts. There is no single pooled headline number.

cohort's own converged runs restores realized false-abort to $\approx$ target ($0.10 \to 0.098$ for the model, 0.100 for RMS@10; Fig. 5, nested-bootstrap split-conformal, model weights frozen). The conformal guarantee is thus a *per-cohort calibration* property, not a transported one; deployment must recalibrate on the target cohort, and the trip-wire reverts to advisory if the realized rate exceeds budget.

Fallback discipline. For BdG, if the Δ channel is unparseable the policy falls back to an rms-only model *with its own* conformal threshold (never comparing an rms-only score to an rms+Δ threshold); both thresholds are certified independently.

7 Deployment evaluation (per-cohort, calibrated)

All numbers are out-of-sample (Replay-OOF Invariant), accounted in SCF iterations (hardware-invariant), and evaluated *on the primary $NbSe_2$-normal cohort with per-cohort recalibration* (Sec. 5). Each abort is charged at the deployment horizon (cumulative iteration 20, when the model first scores under `nsteps`$= 10$).

Recovery and false-abort, ML vs. RMS@10 (Fig. 6, Table 5). With per-cohort recalibration both policies hold realized false-abort at target ($\alpha = 0.05/0.10/0.20 \to 0.04/0.10/0.20$). At matched α the **RMS@10 baseline recovers more wasted compute and catches more doomed runs than the model**: failed-run recall 0.64/0.78/0.86 (RMS@10) vs. 0.53/0.62/0.71 (ML), and fraction of wasted iterations saved 0.61/0.73/0.81 vs. 0.54/0.62/0.69. The model's abort precision stays high (≥ 0.96), but it is the weaker recovery policy on the primary cohort. This is the deployment consequence of the held-out benchmark: *a simple RMS@10 rule, per-cohort calibrated, is the stronger early-abort policy here.*

α economics. Because a false abort under the auto-retry ladder *warm-restarts* (it does not discard a converging run), its cost is small, which pushes the net-optimal α upward. The framework

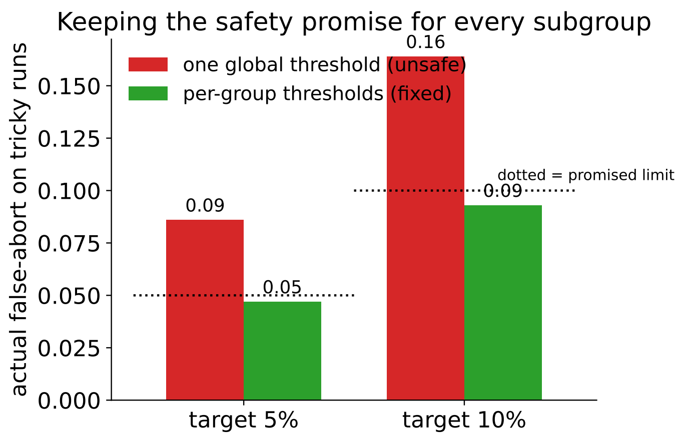

Figure 4: The safety fix. A single global threshold breaks the false-abort promise on the tricky "flat-start" runs (red); one threshold per subgroup (green) restores the target rate within the evaluated strata/cohort.

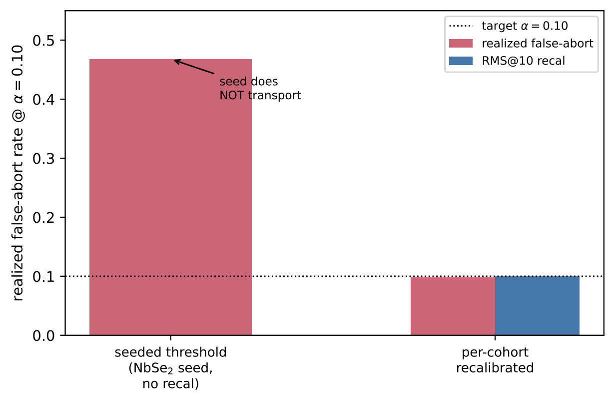

Figure 5: The conformal threshold is a per-cohort object. A $NbSe_2$-seeded threshold applied to held-out data breaks the false-abort promise (0.47 at $\alpha = 0.10$); per-cohort recalibration (weights unchanged) restores realized ≈ target for both the model and RMS@10.

ships two profiles — $\alpha = 0.05$ (conservative) and $\alpha = 0.10$ (throughput) — but the choice of *scorer* (RMS@10 vs. model) is now itself a per-cohort decision, with RMS@10 the default on the primary cohort.

The single-material prospective pilot (narrow). An earlier prospective check calibrated on 329 historical $NbSe_2$ converged scores and applied to a held-out 2026 $NbSe_2$ grid (20 runs, verified absent from training) gave a realized false-abort rate of 0.000 at $\alpha = 0.05$ and 0.10. **This is a narrow, single-material, in-distribution result** and must not be read as broad safety: the same seed, applied to the wider held-out cohort, gave false-abort 0.47 (Sec. 6, Fig. 5). It demonstrates that *material-matched* seeding worked on that single 2026 $NbSe_2$ grid, and reinforces the per-cohort-calibration requirement — not cross-cohort transport.

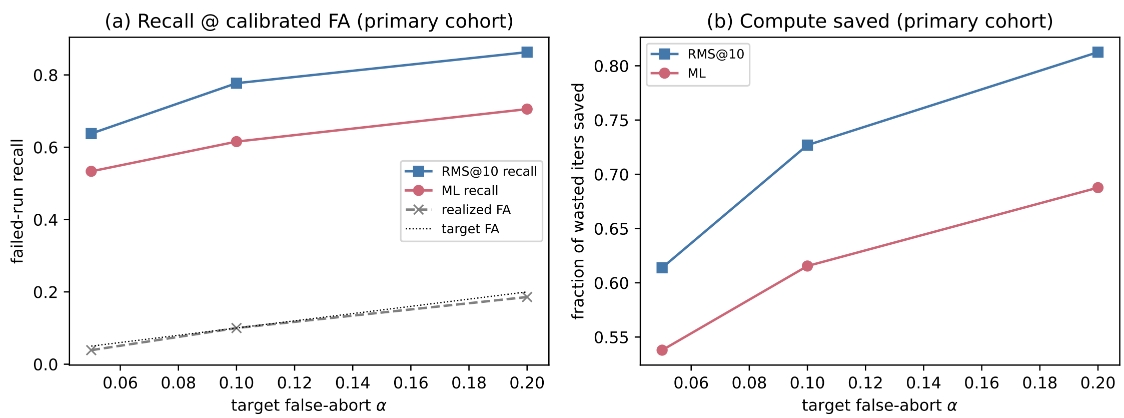

Figure 6: Calibrated risk-savings on the primary $NbSe_2$-normal cohort (per-cohort recalibration, weights frozen). (a) failed-run recall and (b) fraction of wasted iterations saved, both at matched realized false-abort. RMS@10 (blue) dominates the frozen model (red) at every α.

8 Live pilot: the abort machinery works end-to-end

Beyond retrospective replay, we ran the policy *live* in the AiiDA daemon through a staged, gated Fe-doped $NbSe_2$ pilot (Gate 0 → read-only pre-flight → advisory smokes → gated package install/-validation → stock baselines → ml_assist arm). Five systems, each paired against a matched stock counterfactual (same structure, cold start, mixing, segment cadence, partition). The pilot validates the *machinery*, not generalization; at N=5 every effect size is descriptive.

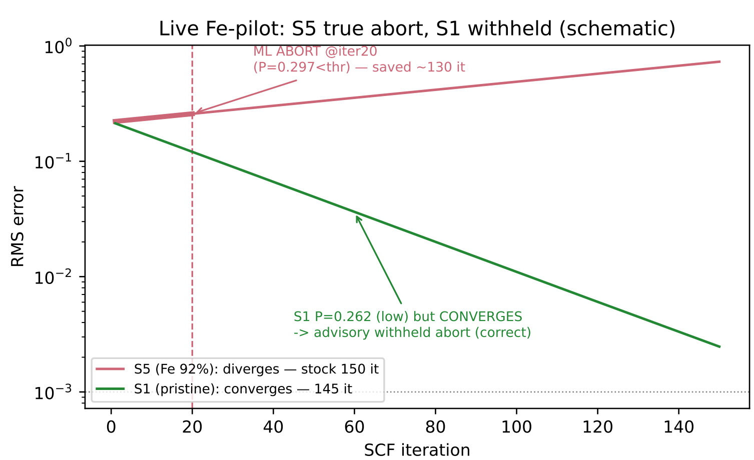

Figure 7: Live pilot exemplars (schematic of recorded runs). **S5** (Fe 92%) diverges; the frozen model scored $P(\text{converge}) = 0.297$ below the abort threshold and issued a live `ERROR_MLASSIST_ABORT` (exit 701) at cumulative iteration 20, saving ≈130 SCF iterations vs. its 150-iteration diverging counterfactual. **S1** (pristine) scored low ($P = 0.262$) yet converges; run advisory, its would-be abort was correctly *withheld* — the deliberate conservative trade-off.

What the pilot shows. (i) A **true early abort** (S5) fired live end-to-end — hook → numpy inference → Mondrian per-stratum threshold → decision → telemetry → exit 701 — and saved ≈130 iterations. (ii) A **would-be false abort was correctly withheld** (S1, advisory), confirming the conservative-advisory default is the right guard for slowly-converging runs. (iii) An **honest miss** (S3): a run whose stall emerges late was not caught by the early window — consistent with the held-out finding that late stalls are the hard case. The abort mechanism, telemetry, and paired accounting are validated; the N=5 deltas are descriptive, not a powered savings claim.

9 The BdG Δ-channel

The standard parser exposes RMS and charge-neutrality but not the pairing amplitude. We wrote a parser for the `l-independent delta` lines KKRhost-BdG prints (two spin components per iteration, before each iteration marker), yielding a per-iteration Δ trajectory (length verified equal to the iteration count). Adding seven Δ-shape features lifts within-BdG *in-distribution* AUC from **0.946 to 0.957** (95% CI on the lift $[+0.000, +0.023]$; Δ-only AUC 0.884). (These are in-distribution 5-fold numbers, like Table 2, not held-out; and BdG runs must be scored with this BdG model — the normal model on BdG data reaches only AUC 0.58, Sec. 5.) The lift is modest here because the historical corpus barely contains pairing-*instability* runs (where Δ grows) — precisely the regime the parser targets for Vehicle A — so it is expected to under-state the value on the campaigns it is built for. The BdG model carries Δ; every Vehicle-A run logs both the rms-only and rms+Δ scores to measure the lift prospectively.

10 Workflow hardening: four BdG footguns and the memory number

A separate but essential outcome: making `kkr_bdg_wc` run a multi-atom cell end-to-end. (Much historical BdG ran instead as standalone `KkrCalculations` in groups; the workchain is the path for *new* BdG runs, and the Δ-parser of Sec. 9 works on any BdG `out_kkr` regardless of how it was launched.) Four defaults, all tuned for a single-atom bulk reference, broke on an 8-atom $NbSe_2$ cell; each was fixed and validated, and the fixes are folded into the workchain (v0.2.1–v0.2.4) and mirrored to the working git branch:

1. **Multi-grid ranks.** MPI ranks must divide *every* step's energy-point grid; a chain with a 24-point normal step and a 32-point BdG step requires a common divisor (or per-step ranks).
2. **Cell-aware `RCLUSTZ`.** The screening-cluster radius now inherits the cell's value instead of a bulk default that overran the compiled cluster limit (`NACLSD`).
3. **NSPIN-aware spin coupling.** BdG with `NSPIN=2` requires coupled spin channels; the workchain now drops the incompatible decoupling flag automatically.
4. **Cell-aware `BZDIVIDE`.** The k-mesh inherits the cell's value; a bulk-dense mesh overran the compiled k-point limit (`KPOIBZ`) once BdG lowered the symmetry.

With these fixed, an 8-atom BdG chain completed for the first time, giving the **Vehicle-A memory number**: BdG job memory is **≈1.0 GB/rank**, *flat with rank count* (energy-parallel replication), i.e. 2.32× the same-cell normal-state memory — a mesh-robust scaling factor used for cluster packing (Fig. 8).

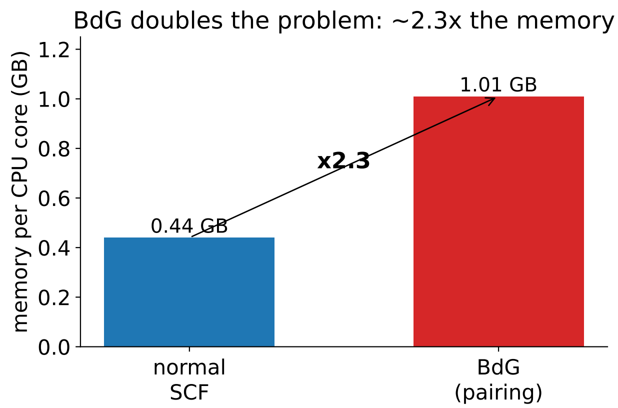

Figure 8: The BdG cost. Doubling the Hamiltonian into Nambu space raises memory per core by $\approx 2.3\times$; the factor is stable across core counts, which sets how many jobs fit on a node.

11 Software package

Everything ships as an installable `aiida-kkr-mlassist` extension (src-layout), with the pure-numpy core cleanly separated from the AiiDA workchain so inference needs no AiiDA and the daemon needs no scikit-learn:

```
aiida_kkr_mlassist/
  features.py canonical 15-D feature schema (train == infer)
  infer.py MLAssistModel - numpy random forest (sklearn-exact)
  conformal.py ConformalAbortPolicy + MondrianConformalAbortPolicy
  ranks.py multi-step MPI-rank / energy-grid validator
  bdg_parser.py BdG Delta-channel parser + features
  decision.py config, decision state-machine, telemetry, graceful failure
  workchain.py MLAssistKkrScfWorkChain (entry point kkr.mlassist)
  models/ frozen artifacts + manifests + MODEL_CARD.md
```

The workchain is opt-in (an absent input reproduces stock behaviour) and records every decision in node extras (advisory shadow data and audit trail). A unit-test suite (feature parity, golden parity, rank validation, conformal shadow-replay, Mondrian control, decision state-machine, BdG parser, dual-scoring) runs under the daemon interpreter to prove the deployment path is scikit-learn–free.

12 Pre-registration and success gates (superseded)

Deployment was originally pre-registered as two campaigns (one shared preamble) with frozen model hashes, the estimand definitions (`TOTAL_CAMPAIGN_ITER`, `MEDIAN_ITER_TO_CONV`), an A/B fairness rule (interleaved within a partition, wall-clock never pooled across partitions), a numeric conformal-violation trip-wire, and a pre-commitment that failures are reported as-is. **This original plan — in particular the Vehicle-B throughput campaign and its *G3* target — was superseded** and is retained here only for provenance:

Vehicle A (BdG, conservative). rms+Δ model, dual-scoring telemetry, $\alpha = 0.05$, Mondrian thresholds; a numeric near-criticality criterion controls how many cells are abort-enabled so the safety rule cannot silently starve the recovery gate; *G2* target: recover $\geq 50\%$ of the stock arm's own waste, zero conformal violations, physics-equality pass. *(Not run — BdG live campaign not launched.)*

Vehicle B (normal, throughput). $\alpha = 0.10$; *paired within-system* A/B (Wilcoxon on iterations) for power; a paired warm-and-cold subset; an explicit racing ladder, commit rule, and warm-start donor-compatibility checks; *G3* target: median iterations-to-converge and total campaign iterations both down $\geq 25\%$.

What actually happened. The full Vehicle-B campaign was *not* run: the planned dopant breadth (Mn/V/Co) lived on the CLAIX cluster, which became unavailable, collapsing the design to an Fe-only iffslurm pilot. That pilot (N=5, Sec. 8) validated the abort *machinery* but is underpowered for a Wilcoxon effect-size test, so **G3 is not an active claim**. The held-out benchmark (Sec. 5) then reshaped the deployment recommendation itself: per-cohort calibration with an RMS@10 comparator, rather than a fixed $\geq 25\%$ throughput gate. Any future campaign must re-pre-register against that recommendation.

Physics-equality tolerances (empirical). Set from the intrinsic numerical reproducibility — the per-run total-energy variation over the last converged iterations (plateau noise, median 1.6×10^{-7} Ry/atom over 2,833 runs). Between-run "rerun" scatter proved unusable (formula-level matching still conflates dopant configurations). The adopted total-energy tolerance is $3\times$ the plateau-noise 90th percentile, $\approx 3 \times 10^{-5}$ Ry/atom — neither contaminated nor pathologically tight.

13 Side studies and honest negatives

Impurity (KKRimp) feasibility. Deferred: 2,390 impurity workchains exist, but their stored RMS trajectory has median length 1 — per-iteration RMS is not in provenance — so early-window prediction cannot be priced without richer logging. A logging change, not a modeling gap.

Warm-start (L3) matched pairs. Underpowered: zero system-matched cold/warm converged pairs exist historically, so the causal warm-start effect is not retrospectively estimable and is deferred to the pre-registered randomized design.

14 Reproducibility and governance

The AiiDA database is treated read-only for analysis; any code-environment or database change is confirmation-gated with a printed plan. Frozen models are versioned artifacts with SHA-256 and a manifest recording feature schema, training-data hash, library versions, and out-of-sample AUC (Appendix B). The Replay-OOF Invariant, the two-denominator reporting rule, and the negatives-are-publishable pre-commitment are the standing methodological safeguards.

15 PotentialBank v1: L3 warm-start realized

The L3 lever — reuse a converged potential instead of a cold Voronoi start — is realized as **PotentialBank v1**: a provenance-clean catalog of certified single-structure converged KKR potentials, each carrying a byte-exact compatibility fingerprint. A donor is admitted only if it passes *E1 certification* verbatim: the parser succeeds and byte-exactly round-trips, the potential's per-type Z-sequence matches the structure site order (empty spheres as Z=0), NATYP equals the site count, the cell is non-CPA and non-BdG, NSPIN/SOC/LMAX/mesh parse, no mapping is guessed, and the staged, AiiDA-retrieved, and manifest SHA-256 all agree. CPA, disordered, and BdG potentials are *excluded by construction.*

Coverage. v1 contains **238 certified donor nodes = 237 unique-SHA donors** (one recorded byte-duplicate: E1 Bi_2X_4 pk 24009≡24011), spanning **8 families, 35 materials, 42 compatibility classes**, of which **33 materials are F2-eligible** (≥ 2 independent certified donors): $NbSe_2$/Bi (E1, 160 donors), simple metals Nb/Al/Cu/Ag/Mo (24), heavy-SOC Au/W/Pt/Sb/Pb (24), magnetic Fe/Ni/Co (13), ordered intermetallics CuZn/Ni_3Al/FeAl (6), covalent empty-sphere Si/Ge (4), the metallic layered chalcogenide $TiSe_2$ (2), and ionic rocksalt MgO/NaCl (4). Each family required its own certified settings recipe (screening cluster RCLUSTZ, Chebyshev log panel R_LOG/NPAN, BZ mesh, SOC key routing, explicit empty-sphere mapping, and a mixing-path variation to obtain independent F2 pairs for deterministic insulator SCF).

Warm-start savings (same-structure). Injecting a certified donor as the start potential cuts iterations-to-converge by large, reproducible margins, *donor-choice-independent* (33/33 targets: identical iterations from either donor) and direction-symmetric where cold-comparable:

family	cold iters	warm iters	% saved
simple metals	66–102	4–7	91–94 %
heavy-SOC	122–136	2–3	∼97 %
magnetic	131–136	5–6	∼96 %
covalent Si/Ge	117–119	2–3	97–98 %
intermetallics	128–152	5–7	95–97 %
$TiSe_2$ (layered)	1188–1354	≤100	∼92 %
ionic MgO/NaCl	171–750	2–500	33–99 %

The magnitude is set by the target's mixing accelerator (a Broyden target polishes a warm start in a few iterations; a pure-linear target still crawls, e.g. 500 vs. 750 for MgO/NaCl), so each row is a *matched same-mixing* warm-vs-cold comparison and we report per-family, never a single pooled headline. Contour caveats are recorded ($TiSe_2$ is coarse-converged only, as the fine contour destabilizes the zero-gap semimetal). Coverage, F2-eligibility, warm-vs-cold, savings distribution, the parked-hard-case map, and the provenance workflow are in Figs. 1–6 of the v1 synthesis package.

Limitations. PotentialBank v1 supports **same-structure reuse only**; it is *not* a cross-material transfer claim and *not* a guaranteed acceleration (donor≡target geometry, chemistry, and settings; savings depend on structure/mixing/contour). CPA/doped/disordered and BdG are out of scope. Nine hard cases are parked with root causes (MoS_2 and NbS_2 NACLSD, WSe_2 SOC contour/EMIN instability, TaS_2 SOC-NaN, TiAl VERTEX3D, Bi_2Te_3/Bi_2Se_3 hard-convergence, CaF_2/LiF). The warm-start outputs are benchmarks and are *not* themselves certified as donors.

16 Status and roadmap

Complete and validated: the framework and its pure-numpy artifact; the conformal (Mondrian) abort policy; the live pilot (abort mechanism validated end-to-end, Sec. 8); the leakage-clean held-out benchmark (Sec. 5); the BdG Δ-parser and frozen BdG models; the `aiida-kkr-mlassist` package with tests; the BdG workchain hardening and memory number; the physics-equality tolerance; the pre-registrations. **Key finding for deployment:** the learned model does not beat a simple RMS@10 rule out-of-sample on the primary cohort, does not transfer across materials, and its conformal guarantee requires *per-cohort recalibration.* Deployment is therefore per-cohort and baseline-benchmarked: the default scorer on the primary cohort is RMS@10, and any campaign must recalibrate the abort threshold on its own converged runs and compare model vs. RMS@10 before enabling abort. **Remaining modeling gap:** late-stall detection (largely missed by the early window, for both the model and RMS@10). **L3 (warm start) is delivered** as PotentialBank v1 (Sec. 15): 237 certified single-structure donors across 8 families with measured same-structure savings; **L2** (mixing search) remains estimated by the pre-registered campaigns. PotentialBank v1 is same-structure reuse only — no cross-material transfer, no guaranteed acceleration.

A Result tables

Table 2: Per-material convergence AUC, *in-distribution* 5-fold cross-validation (deduplicated runs). Strong where data is plentiful; honestly weaker/underpowered where thin. **These are same-distribution CV numbers, not the deployment metric** — for the leakage-clean held-out result see Table 1. No cross-material transfer is claimed.

family	n	OOF AUC	95% CI
$NbSe_2$	373	0.918	[0.86, 0.97]
Bi/NbBi	38	0.849	[0.69, 0.97]
mixed/unknown[2]	1398	0.950	[0.92, 0.97]
pooled (normal, frozen model)	1830	0.944[3]	—

Table 3: The ML margin over baselines, *in-distribution only.* Out-of-sample the relevant baseline is RMS@10 (not the weak extrapolation heuristic), and it matches or beats the model per cohort (Table 1).

method	in-distribution AUC	held-out (primary cohort) AUC
physical extrapolation heuristic	0.549	—
logistic regression	0.826	—
RMS@10 rule (simple baseline)	—	**0.897**
random forest (`ml_assist`)	**0.944**	0.831

Table 4: Marginal vs. Mondrian conformal, false-abort rate (out-of-sample). The marginal threshold under-protects the flat-starter stratum; per-stratum (Mondrian) thresholds restore conditional control.

α	marginal (overall)	marginal (flat-starters)	Mondrian (flat-starters)
0.05	0.049	0.086	**0.047**
0.10	0.098	0.164	**0.093**

[3] In-distribution 5-fold CV; *not* the deployment headline. Out-of-sample on the primary $NbSe_2$-normal cohort the frozen model reaches AUC 0.83 and is beaten by RMS@10 0.90 (Table 1).

[2] "mixed/unknown" = runs whose chemical formula could not be recovered from provenance and so could not be assigned to the $NbSe_2$ or Bi/NbBi families; they are the largest, cleanest group and anchor the pooled model.

Table 5: Calibrated early-abort on the primary $NbSe_2$-normal cohort (held-out, per-cohort recalibration, weights frozen; nested-bootstrap split-conformal, iterations). Realized false-abort holds at target for both; **RMS@10 recovers more and catches more doomed runs than the model at every** α.

α	policy	realized false-abort	failed-run recall	wasted-iters saved
0.05	frozen ML	0.036	0.53	0.54
0.05	RMS@10	0.039	**0.64**	**0.61**
0.10	frozen ML	0.096	0.62	0.62
0.10	RMS@10	0.100	**0.78**	**0.73**
0.20	frozen ML	0.196	0.71	0.69
0.20	RMS@10	0.186	**0.86**	**0.81**

B Model provenance

Frozen artifacts (pure-numpy, golden-parity = 0 vs. scikit-learn, versioned by SHA-256):

model	role	features	in-dist. AUC[4]
`mlassist_v1_normal_fixed10`	normal-state	15	0.944
`mlassist_v1_bdg_rmsdelta`	BdG primary	22	0.957
`mlassist_v1_bdg_rms`	BdG missing-Δ fallback	15	0.946

Each artifact ships a JSON manifest (feature names, training-data SHA-256, library versions, creation timestamp, golden-parity value) and a model card stating intended use, per-system scope, and the Replay-OOF Invariant.

[4]In-distribution 5-fold CV. Held-out, the normal model scores 0.83 on its primary cohort (Table 1); weights were unchanged through all held-out and pilot evaluations.